KARL ANDREI M. MANANGAN

FRONT-END DEVELOPER | UI/UX DESIGNER



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Portfolio: my-portfolio-manangan.vercel.app

PROFESSIONAL SUMMARY

Computer Science student with hands-on experience in front-end development, UI/UX design, and software testing. Led interface design and front-end implementation for Hydro-Scout, a GIS-based fire hydrant monitoring platform for emergency responders. Experienced in building responsive, interactive web interfaces with React.js, Next.js, and Figma.

TECHNICAL SKILLS

Front-end Development:
React.js, Next.js, JavaScript, HTML, CSS, responsive web design

UI/UX Design:
Figma, wireframing, and prototyping

Databases:
Firebase Firestore, Oracle, MySQL, phpMyAdmin

Interactive Mapping:
Mapbox, custom map markers, map interactions

Software Testing:
Selenium IDE, Selenium WebDriver

Other Tools:
Git, GitHub, Python, Jupyter Notebook, Zenmap

PROJECT EXPERIENCE

Hydro-Scout — Hivemind
2026 | Capstone Project

Lead UI/UX Designer & Front-End Developer

- Led the UI/UX design and front-end development of a GIS-based fire hydrant monitoring platform for emergency responders.
- Created wireframes and interactive Figma prototypes to establish the application's layout, navigation, and user interactions.
- Designed and implemented interactive map features using React.js, Next.js, and Mapbox, including custom hydrant pins, information displays, and button behaviors.
- Collaborated with firefighters to understand operational workflows and incorporate user requirements into the interface.

Petshoppe — Hivemind
Academic Team Project

- Collaborated on a full-stack e-commerce pet shop website.

Other Academic Projects

- Automated web testing using Selenium IDE and Selenium WebDriver.
- Developed a restaurant-review sentiment analysis model and applied data mining to retail data.
- Performed network scanning with Zenmap.

EDUCATION

CIIT College of Arts and Technology | 2022-Present
Bachelor of Science in Computer Science

STI College Novaliches | 2020-2022
ICT — Digital Arts

CERTIFICATIONS



